Should Banks Create Money?*

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February 17, 2026

Abstract

How costly are proposals such as 100% reserve narrow banking or a central bank digital currency, which constrain private banks' ability to issue monetary demand deposits? The paper shows that, contrary to widespread concerns, if banks have economically viable alternatives to deposits for funding, constraining bank deposit issuance need not raise loan rates and thus crowd out bank lending. Instead, welfare costs arise from lower interest payments on deposits, which weakens the protection for deposit holders against inflation. A calibration to the U.S. economy suggests, however, that these costs are relatively small, especially when bank reserves are remunerated.

Keywords: *Fractional reserve banking, money supply, narrow banking, CBDC*
JEL codes: E42, E51, G21.

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*The views expressed in this paper do not necessarily reflect those of the Oesterreichische Nationalbank. For comments and discussions I thank Cyril Monnet, Aleksander Berentsen, Chao Gu, V.V. Chari, Chris Phelan, Dirk Niepelt, Timothy Jackson, Lukas Altermatt and Mohammed Ait Lahcen and conference and seminar participants at the EEA-ESEM Conference; Annual MMF Society Conference; Riksbank; Banque de France; Bundesbank; Midwest Macroeconomics Meeting; Oesterreichische Nationalbank; Universities of Minnesota, Torino, Basel and Bern.

1 Introduction

In modern economies, commercial banks supply most of the money held by households and firms under *fractional reserve banking* (FB); that is, they issue digital demand deposits usable for payments in excess of their central bank reserve holdings. In contrast, recent initiatives such as the Swiss "Sovereign Money" referendum of 2018 proposing 100% reserve banking, or the introduction of a retail central bank digital currency (CBDC), would increase the role of banks operating under *narrow banking* (NB), which fully back monetary liabilities with reserves. While 100% reserve banking directly prohibits private money creation, a system with a CBDC constrains FB more indirectly by exposing demand deposits to competition from digital central bank money.[1]

Figure 1: Fractional and 100% reserve banking

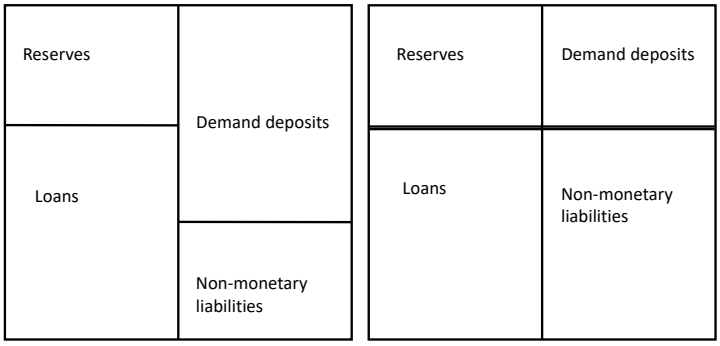


Figure 1 highlights the key difference between FB (left panel) and 100% reserve NB (right panel). Under FB, banks can finance part of their loans with monetary demand deposits, whereas under NB lending must be funded with non-monetary liabilities such as time deposits, bonds, or equity. It has long been understood, and extensively studied since the seminal contribution of Diamond and Dybvig (1983), that the maturity mismatch inherent in lending financed by demand deposits under FB creates the risk of bank runs and financial instability, and that eliminating this mismatch under NB can prevent such runs. Less clear is whether the FB financing structure also provides significant efficiency advantages. While Brunnermeier and Niepelt (2019) and Pennacchi (2012) argue that a carefully designed NB system can replicate any FB allocation, critics such as Ceccetti and Schoenholtz (2014) warn that NB would significantly hamper bank intermediation. In particular, forcing banks to replace demand-deposit funding with more expensive sources such as time deposits is argued to raise loan rates and contract bank lending. These arguments feature prominently in the CBDC discussions (e.g. Andolfatto, 2021; Keister and Sanches, 2022; Chiu et al., 2023) and were also raised during the Swiss referendum debate (e.g. Jordan, 2018; Bacchetta, 2018).

As the financial-stability implications of FB are largely settled, while its efficiency implications remain disputed, the paper focuses on the latter and abstracts from financial stability issues.

[1] Pennacchi (2012) provides a good overview of NB, while Auer et al. (2022) and Infante et al. (2024) survey CBDC proposals. According to Iorio et al. (2024), 81 central banks are currently exploring a CBDC, and 73 run experiments or pilots. How "narrow" the financial system becomes with a CBDC depends on the extent to which non-banks substitute deposits for CBDC. In the extreme, if they fully switch to CBDC, a system with a CBDC is equivalent to a 100% reserve NB system.

The paper aims to clarify the efficiency advantages of FB, or equivalently, the costs of NB, both theoretically and quantitatively, with a particular emphasis on the disintermediation mechanism highlighted above. To this end, I introduce FB, where banks issue monetary deposits subject to a reserve requirement, into the standard New Monetarist framework of Berentsen et al. (2007) (referred to as BCW hereafter). As the paper shows, the allocation in BCW, where banks do not issue monetary deposits and simply reallocate central bank money, is equivalent to the 100% reserve NB allocation of the extended model. The model thus nests the original BCW model as a special case. A key advantage of this framework is that it features distinct demand for both monetary deposits and non-monetary bank liabilities, making it particularly well suited to study the role of funding alternatives to deposits in the disintermediation mechanism described above.[2]
In the model, households acquire money to buy consumption goods, but idiosyncratic liquidity shocks divide them into two groups: savers, who do not need money, and consumers, who need more. Banks, operating under either FB or 100% reserve NB, intermediate between these needs. They extend loans to consumers and finance these loans by issuing monetary deposits to consumers and illiquid, but better-remunerated bonds to savers. These bonds represent the non-monetary bank liabilities shown in Figure 1. As in BCW, banks primarily operate under perfect competition, but I also consider bank market power in the deposit market, following Chiu et al. (2023). Finally, a central bank manages the public money supply through lump-sum transfers by issuing currency to households and potentially interest-bearing reserves to banks, and controls inflation and the nominal interest rate.[3] The central bank sets the nominal interest rate above the Friedman rule and the interest rate on reserves. As a result, holding central bank money is costly for households and banks, which allows bank-issued deposits to circulate as a medium of exchange.
The first main insight of the model is that if banks can issue bonds, the disintermediation mechanism highlighted above, where NB lowers bank lending through higher loan rates, does not apply. Although, as illustrated in Figure 1, loan financing shifts from deposits to more expensive bonds with NB, this does not increase the loan rate. The same holds if the central bank issues an interest-bearing CBDC. Bonds are crucial for this result because they provide insurance against the liquidity shock: after a shock, savers are left with idle money balances. Illiquid bonds offering a higher return than deposits are thus an attractive savings vehicle. In equilibrium, perfect competition in the bond market drives the bond return up to the nominal interest rate. As a result, bonds offer full insurance against liquidity shocks for savers, and households more broadly. This outcome holds under both FB and NB, since banks issue bonds regardless of whether deposit issuance is constrained. Furthermore, because bonds are so valuable to savers, it is the savers (bond holders), not the consumers (deposit holders), who determine the loan rate. This isolates the loan rate from constraints in deposit issuance. The model thus rationalizes the idea that introducing 100% reserve NB or a CBDC may not significantly affect loan rates because financial intermediaries can finance

[2]Funding alternatives to deposits are either absent or play no essential role in many existing models. In Chiu et al. (2023), for example, although banks can in principle finance lending with non-monetary liabilities, this option is not used in equilibrium, and in Keister and Sanches (2022) and Andolfatto (2021) banks cannot issue other liabilities than deposits. Infante et al. (2024) note the general scarcity of research on this topic.

[3]In Section 4.3, I also consider a model where households have access to an interest-bearing CBDC.

their assets using alternative, less liquid liabilities than deposits. This confirms similar results from partial equilibrium models such as Whited et al. (2023).

The second insight is that NB can still be costly because it reduces the interest paid on deposits. The mechanism is as follows: as highlighted in Figure 1, under FB, deposits finance both loans and reserves, while under NB, deposits only fund reserves. Since loans yield a higher return than reserves in equilibrium, banks earn more on deposit-financed assets under FB than under NB. When competition is sufficiently strong, this higher income is passed on to deposit holders in the form of higher interest payments. This is economically valuable because it helps to protect deposit holders from the inflation tax, especially when inflation and nominal interest rates are high.

To quantify this effect, I calibrate the model to the U.S. economy and compute the welfare cost of lower deposit rates under NB. When reserves are unremunerated, the welfare cost of 100% reserve NB, measured as the income households must give up under FB to reach the same welfare level as under NB, amounts to 0.03% of GDP at a nominal interest rate of 5%. At nominal rates of 10% and 15%, the cost rises to 0.29% and 0.61% of GDP, respectively. These costs fall sharply when reserves are remunerated. For instance, at a 15% nominal interest rate, the welfare cost falls from 0.61% of GDP to between 0.02% and 0.03%, depending on how closely reserve remuneration follows the nominal interest rate. Reserve remuneration is crucial, since under NB, where deposits only fund reserves, banks can pay interest on deposits only if those reserves earn interest.[4]

In sum, the paper suggests that if banks have economically meaningful alternatives to monetary deposits for funding, the costs of NB are relatively small, especially if reserves are remunerated. To the extent that NB reduces the inherent fragility of FB, which is plausible at least within the payment system, this implies that enhancing financial stability with NB may not be so costly.[5]

The rest of the paper is organized as follows. Section 2 introduces the environment. Section 3 presents the baseline model with perfect competition and extends it to incorporate imperfect competition. Section 4 calibrates the model with imperfect competition to the U.S. economy and quantifies the welfare costs of 100% reserve NB, as well as the effects of an interest-bearing CBDC. Section 5 concludes.

2 Environment

The environment builds on Lagos and Wright (2005) and in particular on BCW. Time is discrete and continues forever. There are three types of agents: a continuum of households of measure 1, a continuum of commercial banks of measure 1 (hereafter, banks),[6] and a central bank. Each period is divided into two sequential, competitive markets, called first market (FM) and second market

[4]Because the central bank implements changes in the money supply through lump-sum transfers, reserve remuneration and inflation are independent in the model. As a result, paying interest on reserves does not generate distortions via higher inflation. I thank an anonymous referee for this insight.

[5]Current FB systems include various regulations to address financial instability, most notably deposit insurance. Ultimately, the costs and benefits of NB regulations should thus be weighed against those of deposit insurance and other financial stability regulations. Jackson and Pennacchi (2021) take important steps in this direction.

[6]In Section 3.5, with imperfect competition, there will be a finite number N of banks.

(SM).[7] In each market, a perishable consumption good is traded, denoted by q in the FM and x in the SM. Households are infinitely lived and discount future periods at rate β. At the beginning of each period, they face a preference shock that divides them into two groups. With probability $s \in (0,1)$, a household becomes a seller in the FM and produces with linear disutility $c(q_s) = q_s$. With the complementary probability, the household becomes a buyer and consumes in the FM with utility $u(q_b)$, where $u'(0) = \infty$, $u'(q_b) > 0$, and $u''(q_b) < 0$. Let buyers' relative risk aversion be given by $\sigma = -\frac{q_b u''(q_b)}{u'(q_b)} < 1$, which is assumed to be constant and less than one.[8] In the SM, all households consume x and produce h, with quasi-linear utility $U(x) - h$, where $U'(0) = \infty$, $U'(x) > 0$, and $U''(x) < 0$. Households lack commitment and cannot enforce debt repayments. As a result, buyers cannot issue debt to purchase goods from sellers in the FM, and households find it useful to hold a means of payment, i.e., money.

Before describing the money supply by banks and the central bank, we first derive the first-best allocation. With perishable consumption goods, a social planner chooses non-negative consumption and production levels to maximize the expected per-period welfare of a household, subject to feasibility constraints:

$$\max_{q_b,x_b,h_b,q_s,x_s,h_s} W = (1-s)\left[u(q_b) + U(x_b) - h_b\right] + s\left[-c(q_s) + U(x_s) - h_s\right], \tag{1}$$

$$s.t. \quad (1-s)q_b = sq_s, \tag{2}$$

$$(1-s)x_b + sx_s = (1-s)h_b + sh_s. \tag{3}$$

Constraints (2) and (3) ensure that aggregate consumption equals aggregate production in the FM and SM. The efficient quantities q_b^*, q_s^*, and $x_b^* = x_s^* = x^*$ then equalize the marginal utility of consumption and the marginal disutility of production in both markets:

$$\frac{u'(q_b^*)}{c'(\frac{1-s}{s}q_b^*)} = 1, \quad q_s^* = \frac{1-s}{s}q_b^*, \quad U'(x^*) = 1. \tag{4}$$

The central bank issues fiat money M, referred to as reserves, which can be held by both households and banks, and grows at rate $\gamma = M/M_{-1}$.[9] Reserves held by banks i_r are remunerated at rate $i_r \geq 0$, while reserves held by households i_z are unremunerated, i.e., $i_z = 0$. This reflects the current monetary system in which only commercial banks have access to interest-bearing central bank money, while non-banks may only hold unremunerated currency.[10] The central bank manages the supply of reserves through lump-sum transfers τ to households in the SM. Accordingly, seigniorage

[7] Since both markets are competitive, I follow BCW and refer to them as first and second market, instead of the more common terms decentralized and centralized market used in the New Monetarist literature.

[8] This condition ensures that real deposit holdings increase with the deposit rate, so banks face a trade-off when issuing deposits under imperfect competition. The calibration yields $\sigma = 0.19$, supporting this assumption empirically.

[9] For brevity, variables referring to the representative period t are written without a subscript, those from the previous period use -1, and those from the following period use $+1$. The terms "money" and "reserves" are used interchangeably.

[10] The European Central Bank has remunerated bank reserves since its establishment in 1999, while the Federal

revenue in period t, $M - M_{-1}$, which may be negative if the money supply contracts, is used to finance transfers τ and interest payments on reserves.

Let $1/\phi$ denote the price of x in terms of reserves in the SM, and let p denote the price of q in terms of reserves in the following SM. Define inflation as ϕ_{-1}/ϕ. Under stationarity, the real value of money remains constant, i.e., $\phi M = \phi_{-1} M_{-1}$, and the money growth rate γ equals the inflation rate. The central bank therefore perfectly controls long-run inflation and the nominal interest rate, which is defined as $1 + i = \gamma/\beta$, the product of inflation γ and the real interest rate $1/\beta$. Throughout the paper I assume that holding reserves is costly, i.e., the opportunity cost for holding money, i, exceeds both the return on reserves for households ($i_z = 0$) and the return on reserves for banks (i_r):

$$i > i_z \quad , \quad i > i_r. \tag{5}$$

Banks are risk-neutral firms owned by households. They can commit to and enforce debt contracts at no cost, enabling them to extend loans and issue debt. Specifically, banks offer loans at interest rate i_l, they issue deposits at rate i_d, and they issue bonds at rate i_b. Deposits, which are subject to a reserve requirement $\alpha \in (0, 1]$,[11] serve as a second means of payment alongside reserves in the FM, while bonds are not usable in these transactions. Bonds thus represent the non-monetary bank liabilities referred to in the introduction.[12] All financial contracts are nominal, formed after the preference shock in the FM, and fully redeemed in the subsequent SM.[13] Because banks interact with households after the preference shock, they intermediate between sellers, who do not need liquidity in the FM, and buyers, who do. Typically, buyers borrow liquidity (deposits or reserves) from banks, and sellers convert their liquid balances into illiquid but better-remunerated bonds. As a result, loan and deposit demand are closely linked on the buyer side. This contrasts with Chiu et al. (2023), Keister and Sanches (2022), or Andolfatto (2021), where borrowers are entrepreneurs, and loan and deposit demand are independent. As a baseline, the paper first considers competitive banking. In Section 3.5, I then introduce imperfect competition, where a fixed number N of banks compete à la Cournot in the deposit market, following Chiu et al. (2023).

Reserve began paying interest on bank reserves in October 2008. To explore the implications of an interest-bearing retail CBDC, I introduce remunerated reserves for households ($i_z \geq 0$) in Section 4.3.

[11] The reserve requirement can be interpreted either as a regulatory or a technological constraint. While the former may be set to zero (for example, Canada or the UK operate without formal minimum reserve requirement), the latter is strictly positive. Even in the absence of a regulatory requirement, banks still need central bank reserves to settle the transactions made with deposits. This justifies assuming $\alpha > 0$.

[12] In BCW, banks only issue illiquid liabilities and play no role in the money supply. The introduction of liquid bank liabilities (deposits) subject to a reserve requirement is therefore the main extension relative to BCW.

[13] With quasi-linear utility in the SM, there is no welfare gain from spreading loan repayment or debt redemption across multiple periods. Thus, restricting contracts to this form is not a constraint in this environment.

3 The Model

3.1 Banks

Banks acquire reserves r and loans l by issuing deposits d and bonds b. They maximize profits π, subject to the reserve requirement and a balance sheet constraint:

$$\max_{\{l,r,d,b\}\geq 0} \pi = r(1+i_r) + l(1+i_l) - d(1+i_d) - b(1+i_b), \tag{6}$$
$$s.t. \quad \alpha d \leq r \quad , \quad l + r = d + b.$$

We focus on the case with positive loans and conjecture that in equilibrium $i_l \geq i_d$ and $i_l > i_r$, i.e., financing loans with deposits is profitable, and loans yield a higher return than reserves. Under these assumptions, the reserve requirement binds and the objective function can be rewritten as:

$$\max_{d,b} \; d\left(\alpha i_r + (1-\alpha) i_l - i_d\right) + b\left(i_l - i_b\right).$$

Any equilibrium with positive household demand for both deposits and bonds therefore requires:

$$i_d = \alpha i_r + (1-\alpha) i_l, \tag{7}$$
$$i_b = i_l, \tag{8}$$

and zero profits for banks. The deposit rate is a weighted average of the returns on deposit-financed assets, i_r and i_l. Thus under FB $i_d \in (i_r, i_l)$, while under NB with $\alpha = 1$, we have $i_d = i_r$ because deposits are only backed by reserves. Because $i_l > i_r$ in equilibrium, the following interest rate relationships hold, consistent with the assumptions above:

$$i_l = i_b > i_d \geq i_r. \tag{9}$$

Because the reserve requirement binds, it also determines the monetary regime. If $\alpha < 1$, banks issue more deposits than they hold in reserves, characterizing a FB system. If $\alpha = 1$, deposits are fully backed by reserves, defining a NB system.

3.2 Buyers

Let $V_b(m)$ denote the value of a household who enters period t with m units of reserves and has just learned that she is a buyer. Also, let $V(m_{b+1})$ denote the expected value of a buyer entering period $t+1$ with m_{b+1} units of reserves. The buyer's problem is given by:

$$V_b(m) = \max_{\{q_b, x_b, h_b, m_{b+1}, l_b, d_b, m_b, b_b\} \geq 0} \quad u(q_b) - h_b + U(x_b) + \beta V(m_{b+1})$$

$$s.t. \quad b_b + m_b + d_b \leq m + l_b, \tag{10}$$

$$pq_b \leq m_b + d_b(1 + i_d), \tag{11}$$

$$h_b + \phi\left(m_b + d_b(1 + i_d) - pq_b\right) + \phi(\tau + \pi) + b_b\phi(1 + i_b) \tag{12}$$
$$= x_b + \phi m_{b+1} + l_b\phi(1 + i_l).$$

After the preference shock, buyers first acquire financial assets and borrow from banks. Using their reserves m and loan amount l_b, they acquire bonds b_b, reserves m_b, and deposits d_b, as shown in (10). In the FM, buyers use either reserves or deposits to purchase goods from sellers, as described in (11).[14] When buyers enter the SM, they receive income from labor h_b, from monetary wealth left over after the FM $m_b + d_b(1 + i_d) - pq_b$, from central bank transfers and bank profits $\phi(\tau + \pi)$, and from bond holdings $b_b\phi(1 + i_b)$. These resources are used to purchase consumption goods x_b, acquire new money holdings ϕm_{b+1}, and repay loans $l_b\phi(1 + i_l)$, as shown in (12).

Let δ denote the Lagrange multiplier on (10) and λ the multiplier on (11). Substituting (12) into the objective function, the buyer's problem yields the following first-order conditions, assuming interior solutions for q_b, x_b, m_{b+1} and l_b:

$$q_b : \qquad u'(q_b) = p\left(\phi + \lambda\right), \tag{13}$$

$$x_b : \qquad U'(x_b) = 1, \tag{14}$$

$$m_{b+1} : \qquad \beta V'(m_{b+1}) = \phi, \tag{15}$$

$$l_b : \qquad \delta = \phi(1 + i_l), \tag{16}$$

$$d_b : \qquad (1 + i_d)(\phi + \lambda) \leq \delta, \tag{17}$$

$$m_b : \qquad \phi + \lambda \leq \delta, \tag{18}$$

$$b_b : \qquad \phi(1 + i_b) \leq \delta. \tag{19}$$

To consume in the FM, buyers must hold either reserves or deposits. Since deposits are interest-bearing and reserves are not (for households), the marginal benefit of holding deposits, given by the left-hand side of (17), exceeds that of holding reserves, as shown in the left-hand side of (18). As a result, buyers strictly prefer to hold deposits and (17) binds, while (18) is slack. Combining (17) and (16), the multiplier of (11) then reads:

$$\lambda = \phi\left(\frac{1 + i_l}{1 + i_d} - 1\right).$$

[14] Note that p denotes the price of q in terms of reserves in the *next* SM. This is why the interest rate i_d enters the budget constraint. One can interpret $m_b + d_b(1 + i_d)$ as the buyer's "monetary wealth" of a buyer in the subsequent SM.

Thus, as long as $i_l > i_d$ (which follows from the bank problem), $\lambda > 0$ and (11) binds, i.e., buyers use all of their deposits in the FM. Since the bond rate equals the loan rate, buyers could in principle increase borrowing and to acquire bonds. However, because this additional borrowing/bond holding does not affect the real allocation, I assume buyers do not hold bonds. Finally, from (14), buyers consume the efficient quantity in the SM, and from (15), the expected marginal value of new reserve holdings equals their marginal cost. Summarizing, the solution to the buyer problem is characterized by:

$$u'(q_b) = p\phi\frac{1+i_l}{1+i_d}, \tag{20}$$

$$x_b = x^*, \tag{21}$$

$$\beta V'(m_{b+1}) = \phi, \tag{22}$$

$$d_b = \frac{pq_b}{1+i_d}, \tag{23}$$

$$l_b = d_b - m, \tag{24}$$

$$m_b = 0 \quad , \quad b_b = 0. \tag{25}$$

This also yields the envelope condition to the buyer problem:

$$V_b'(m) = \delta = \phi(1+i_l). \tag{26}$$

3.3 Sellers

The problem for sellers is analogous to that of buyers, except for constraint (11), which is absent since sellers do not consume in the FM.

$$\begin{aligned} V_s(m) = \max_{\{q_s,x_s,h_s,m_{s+1},l_s,d_s,m_s,b_s\}\geq 0} \quad & -c(q_s) - h_s + U(x_s) + \beta V(m_{s+1}) \\ s.t. \quad & b_s + m_s + d_s \leq m + l_s, \\ & h_s + \phi\left(m_s + d_s(1+i_d) + pq_s\right) + \phi(\tau+\pi) + b_s\phi(1+i_b) \\ & = x_s + \phi m_{s+1} + l_s\phi(1+i_l). \end{aligned}$$

As before, I substitute h_s in the objective function using the seller's budget constraint.[15] Assuming interior solutions for q_s, x_s, and m_{s+1}, we obtain the following first-order conditions:

[15] This assumes $h_s > 0$, which requires a scaling condition on $U(.)$. As we will be shown, sellers also consume the efficient quantity in the SM, $x_s = x^*$. Hence, $h_s > 0$ holds if x^* is sufficiently large.

$$
\begin{aligned}
q_s: &\quad p\phi = c'(q_s), \\
x_s: &\quad U'(x_s) = 1, && (27)\\
m_{s+1}: &\quad \beta V'(m_{s+1}) = \phi, && (28)\\
l_s: &\quad \delta \leq \phi(1+i), \\
d_s: &\quad (1+i_d)\phi \leq \delta, \\
m_s: &\quad \phi \leq \delta, \\
b_s: &\quad \phi(1+i_b) \leq \delta. && (29)
\end{aligned}
$$

Since sellers do not require liquidity in the FM, their choice of financial assets is driven solely by return considerations. From the bank's problem and equation (9), we know that $i_b > i_d$ and $i_b > i_r$. Therefore, sellers prefer to hold bonds over deposits and reserves, equation (29) holds with equality, and $d_s = m_s = 0$. Because $i_b = i_l$, sellers are indifferent to borrowing additional funds to acquire more bonds. As with buyers, I exclude this type of borrowing from the analysis. According to (27), sellers also consume the efficient quantity in the SM. Furthermore, their first-order condition for reserve holdings, (28), is identical to that of buyers in (15). In particular, this condition does not depend on the household's type or wealth in the SM. As a result, buyers and sellers choose the same reserve holdings in the SM, i.e., $m_{b+1} = m_{s+1} = m_{+1}$. This is an implication of the quasi-linear utility structure introduced by Lagos and Wright (2005). We can summarize the solution to the seller's problem as follows:

$$
\begin{aligned}
&c'(q_s) = p\phi, && (30)\\
&x_s = x^*, && (31)\\
&\beta V'(m_{+1}) = \phi, && (32)\\
&d_s = l_s = m_s = 0, && (33)\\
&b_s = m. && (34)
\end{aligned}
$$

The envelope condition with respect to reserve holdings is:

$$
V_s'(m) = \delta = \phi(1+i_b). \tag{35}
$$

3.4 Equilibrium

We focus on stationary equilibria in which real allocations remain constant over time. We first derive households' optimal reserve holdings in the SM and all market clearing conditions. By iterating (35) and the envelope condition of buyers, (26), one period forward, the optimality condition for households' reserve holdings in the SM can be rewritten as:

$$\begin{aligned}\phi = \beta V'(m_{+1}) &= \beta s V_s'(m_{+1}) + \beta(1-s)V_b'(m_{+1}) \\ &= \beta s \phi_{+1}(1+i_{b+1}) + \beta(1-s)\phi_{+1}(1+i_{l+1}).\end{aligned} \tag{36}$$

The marginal cost of holding reserves, ϕ, equals the marginal expected benefit for sellers and buyers. For sellers, the marginal benefit is the real bond return, since they acquire bonds. For buyers, it is the real loan rate, as bringing more reserves reduces the amount they need to borrow, thereby saving on interest payments. Market clearing for reserves in the SM is given by:

$$(1-s)m_{b+1} + sm_{s+1} = m_{+1} = M. \tag{37}$$

Suppose that banks in total supply D deposits, B bonds, and hold L loans and R reserves during their interaction with households before the FM. Market clearing conditions for reserves, bonds, deposits, and loans are thus given by:

$$R + (1-s)m_b + sm_s = R = M_{-1}, \tag{38}$$

$$(1-s)b_b + sb_s = sM_{-1} = B, \tag{39}$$

$$(1-s)d_b = D = \frac{R}{\alpha} = \frac{M_{-1}}{\alpha}, \tag{40}$$

$$(1-s)l_b = L = D + B - R = \left(\frac{1-\alpha}{\alpha} + s\right) M_{-1}. \tag{41}$$

Equation (38) implies that total reserves held by banks R, plus those held by households, must equal the total reserves brought into the period by households, M_{-1}. Since $m_b = m_s = 0$, banks hold all reserves. In the bond market (39), only sellers hold bonds, which must equal total supply B issued by banks. In the deposit market (40), only buyers hold deposits. Due to the binding reserve requirement, total deposit supply equals $R/\alpha = M_{-1}/\alpha$, since banks hold all the reserves by (38). In the loan market (41), buyer loan demand equals bank loan supply $D+B-R$. Note that all the market clearing conditions can be expressed as a linear function of total reserves. Finally, the FM and the SM clear according to (2) and (3).

To solve for the equilibrium interest rates, we combine the condition for optimal reserve holdings (36) with the interest rate relationships derived from the bank problem, (7) and (8), and impose stationarity, where $\gamma = \phi/\phi_{+1}$. This yields:

$$i_l = i_b = i, \tag{42}$$

$$i_d = (1-\alpha)i + \alpha i_r. \tag{43}$$

In equilibrium, the loan rate and the bond rate both equal the nominal interest rat. As conjectured in the banking problem, $i_l > i_r$ and $i_l > i_d$ hold. Note that the deposit rate decreases with the reserve requirement, and is therefore higher under FB than under NB.

To derive equilibrium FM consumption, combine buyers' optimal consumption condition (20) with sellers' optimal production condition (30), and substitute the expressions for equilibrium interest rates along with the FM market-clearing condition (2):

$$\frac{u'(q_b)}{c'(\frac{1-s}{s}q_b)} = \frac{1+i}{1+i_d}. \tag{44}$$

The rest of the equilibrium allocation is given by:

$$\begin{aligned}
q_s &= \frac{1-s}{s}q_b, \\
\phi D = (1-s)q_b\frac{c'(q_s)}{1+i_d} \quad &, \quad \phi M_{-1} = \alpha\phi D \quad , \quad \phi L = (1-\alpha+\alpha s)\,\phi D \quad , \quad \phi B = s\alpha\phi D, \qquad (45) \\
x_b &= x_s = x^*, \\
h_s &= x^* + \phi M_{-1} - \frac{\phi D(1+i_d)}{s} - \phi M_{-1}(1+i), \qquad (46) \\
h_b &= x^* + \phi M_{-1} + \frac{\phi L(1+i)}{1-s}. \qquad (47)
\end{aligned}$$

Since consumption in the SM is always efficient at x^*, we can focus on the FM allocation in (44) for welfare analysis. Note that q_b in (44) decreases with the spread between i and i_d, and without spread achieves the efficient allocation q_b^* from (4).[16] Therefore the welfare analysis crucially depends on the deposit rate i_d. Proposition 1 and Figure 2 summarize the main effects of the reserve requirement α on the equilibrium allocation.

Proposition 1. *Suppose holding money is costly, the banking system is fully competitive, and either operates under FB with $\alpha \in (0,1)$, or under NB with $\alpha = 1$. Then there exists a stationary equilibrium in which:*

a) Banks' ability to insure households against preference shocks and the loan rate are independent of α.

b) The deposit rate, deposit and loan quantity, consumption and welfare all decrease in α.

c) The NB allocation ($\alpha = 1$) is equivalent to the allocation in BCW.

d) As $\alpha \to 0$ the allocation converges to the first-best.

To better understand Proposition 1, recall that the fundamental source of inefficiency in the baseline model is that holding central bank reserves is costly, i.e., the central bank sets a positive i above i_r. As a result, households hold too few reserves relative to the social optimum, and inflation acts like a tax on real activity. The preference shock further amplifies this basic inefficiency. The risk of becoming a seller with idle, costly liquidity makes acquiring money even less attractive ex ante. Banks play two roles in this environment. First, they provide an alternative liquid asset to reserves (deposits), that yields a higher return and thus better compensates the deposit holders against the

[16] From the strict concavity of $u(q)$, the left-hand side of (44) decreases in q_b, so a higher spread implies a lower q_b.

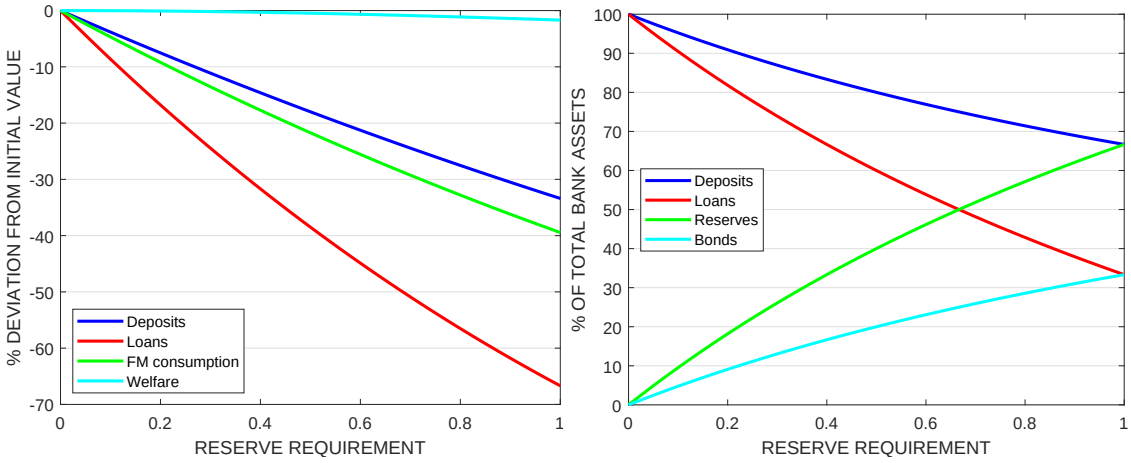
inflation tax. Second, they intermediate liquidity from sellers to buyers after the preference shock, i.e., they provide insurance against liquidity shocks.

As Proposition 1a) highlights, the reserve requirement does not affect banks' second role. The reason is that banks provide liquidity insurance through bonds, which they issue independently of constraints on deposit issuance. To see this, note that for sellers, illiquid bonds offering a higher return than deposits serve as a useful savings vehicle. In equilibrium, since $i_b = i$, sellers who acquire bonds are fully compensated for the costs of holding money. Liquidity insurance is therefore perfect under FB and under NB, and the preference shock s does not affect q_b in (44). Since bonds are so valuable for sellers, it follows also that it is the bond holders (sellers) and not the deposit holders (buyers) who determine the loan rate, as shown in (36). As a result, i_l is independent of changes in i_d over α. Thus, even though banks replace deposit-financed lending with bond-financed lending as α increases, this shift does not raise i_l and thus reduce lending.[17] This changes if banks are unable to issue bonds. In this case, as appendix A.2 shows, i_l increases with α, and a higher α also worsens liquidity insurance. In sum, Proposition 1a) thus emphasizes the importance of funding alternatives to deposits for banks.

Proposition 1b) also highlights however, that the reserve requirement negatively affects the first role of banks, namely, their ability to provide a means of payment (deposits) that offers a higher return than reserves. The reason is straightforward: as α increases, banks can finance a smaller share of loans with deposits. And since the return on loans exceeds the return on reserves, this lowers banks' income on deposit-financed assets. Under perfect competition, this directly translates into a lower deposit rate, i_d. As a result, deposits become less attractive to buyers, they borrow and consume less, and welfare declines, as shown in the left panel of Figure 2.

[17] Intuitively, (36), which equalizes $i_l = i$ in equilibrium, implies a vertical deposit and loan supply. To see this, note that if $i_l < i$, (36) implies that $\phi M_{-1} = 0$. If the marginal return of holding reserves does not cover the marginal costs, households will not hold reserves. But in that case total bonds, deposits and loans, which are all proportional to reserves by the bank balance sheet identities, must be zero too. Only at $i_l = i$ are households willing to hold reserves, and banks can provide bonds, deposits and loans. As a result, if α increases and buyers' deposit and loan demand shifts down, this keeps the loan rate unaffected.

Figure 2: Effects of the Reserve Requirement on the Equilibrium Allocation



Notes: Based on the utility functions and calibrated parameters from Section 4, evaluated at $i = 10\%$ and $i_r = 0$. Since the allocation at $\alpha = 0$ is efficient, the left panel of Figure 2 reports the percentage deviation from the first-best allocation.

In the extreme case, namely, a NB economy with $\alpha = 1$, where banks fully back deposits with reserves, we have $i_d = i_r$, and the advantage of banks to better compensate deposit holders disappears. The right panel of Figure 2 shows that as $\alpha \to 1$, and deposits converge to reserves, loans become fully financed by bonds. As Proposition 1c) highlights, the NB allocation is equivalent to an economy in which banks do not issue monetary liabilities and instead reallocate central bank reserves after the preference shock, as in BCW.[18] Intuitively, an economy with only central bank money is equivalent to an economy where banks issue liquid liabilities fully backed by central bank reserves, as under NB. Either way, commercial banks play no essential role in the money supply.
In the other extreme case, when banks create deposits and make loans without constraint ($\alpha = 0$), the spread between i_d and i_l vanishes, and banks perfectly compensate deposit holders for inflation. This yields the efficient allocation (4). As shown in the right panel of Figure 2, in this equilibrium banks do not hold reserves, and central bank money is not valued. Intuitively, by issuing deposits fully backed by loans, banks fully compensate for the lack of commitment by households and implement the direct credit allocation between buyers and sellers. The right panel of Figure 2 also shows that when $\alpha = 0$, banks do not issue bonds. Without reserve requirement, deposits have no disadvantage relative to bonds, and bonds become inessential.

3.5 Imperfect Competition

Following Chiu et al. (2023), I assume Cournot competition among N banks in the deposit market, while the bond and loan markets remain perfectly competitive. This implies that each individual bank j internalizes the effect of its own deposit issuance on the deposit rate through market clearing, taking the deposit supply of all other banks $i \neq j$ as given. However, it ignores any similar effects

[18] In footnote 9 of BCW, the authors also interpret their model as a NB economy.

on the loan and bond rates. From the perspective of bank j, total real deposit supply is given by:

$$\phi D = \phi \left(\sum_{i \neq j}^{N} d_i + d_j \right) \tag{48}$$

where d_j denotes the bank's own deposit issuance, and $\sum_{i \neq j}^{N} d_i$ is the aggregate supply from all other banks. Bank j anticipates total real deposit demand from buyers using conditions (20) and (23) as:

$$\phi D = \begin{cases} 0 & \text{if} \quad i_d < 0, \\ (1-s)\dfrac{q_b u'(q_b)}{1+i_l} & \text{if} \quad i_d \geq 0. \end{cases} \tag{49}$$

Buyers demand deposits only if the deposit rate is non-negative; otherwise, they switch to holding reserves, and the demand for deposits is zero.

Equalizing (48) and (49) yields a relationship between the deposits issued by bank j and the deposit rate, which I denote by $i_d(\phi D)$, where ϕD is defined in (48). The economically interesting case arises when real deposit demand (49) increases with the deposit rate, i.e., when $\partial \phi D / \partial i_d > 0$ and therefore $\partial i_d(\phi D)/\partial \phi D > 0$. In this case, if the bank issues more deposits, buyers must be compensated with a higher return to be willing to hold the additional deposits. Only then does the bank face a trade-off when issuing deposits.[19] From (49), real deposit demand increases in the deposit rate if σ, the coefficient of relative risk aversion, is less than one, an assumption made in the environment.[20]

The problem of the representative bank j is similar to the case under perfect competition, as described in (6). Again, when $i_l \geq i_d$ and $i_l > i_r$, the reserve requirement binds, and the bank's problem becomes:

$$\max_{d_j, b_j} \quad d_j \left(\alpha i_r + (1-\alpha) i_l - i_d(\phi D) \right) + b_j \left(i_l - i_b \right),$$

where the first-order condition for deposit issuance is:

$$(1-\alpha) i_l + \alpha i_r - i_d = i_d^{PC} - i_d = \phi d_j \frac{\partial i_d(\phi D)}{\partial \phi D}, \tag{50}$$

and i_d^{PC} denotes the deposit rate under perfect competition. Under Cournot competition, the bank internalizes that the deposit rate depends on its own issuance, which drives i_d below i_d^{PC}.[21]

We focus on a symmetric pure strategy equilibrium in which all banks issue the same amount of deposits, i.e., $d_j = d_i = d = D/N$. Under this assumption, (50) simplifies to a familiar expression

[19] In the opposite case, i.e., if a bank could issue more deposits and buyers were still willing to hold them at a lower deposit rate, the bank would always issue as many deposits as possible, driving i_d down to its minimum, $i_d = 0$.

[20] The calibration yields $\sigma = 0.19$, consistent with this assumption.

[21] The resulting spread implies that banks earn positive profits in equilibrium. These profits are distributed to households in the SM, which does not affect the allocation as long as sellers continue to work in the SM; see also footnote 15.

equating a Lerner-type index with the inverse interest elasticity of real deposit demand, scaled by the number of banks N (Freixas and Rochet, 2008, p. 78–80):

$$\frac{i_d^{PC} - i_d}{i_d} = \frac{1}{\varepsilon_D(i_d)} \frac{1}{N} \quad \text{where} \quad \varepsilon_D(i_d) = \frac{i_d}{\phi D} \frac{\partial \phi D}{\partial i_d}. \tag{51}$$

Equation (51) implies that the spread $i_d^{PC} - i_d$ converges to zero either as the banking sector becomes perfectly competitive (i.e., $N \to \infty$), or as deposit demand becomes perfectly elastic (i.e., $\varepsilon_D \to \infty$). Denoting the solution of (51) by i_d^*, the equilibrium deposit rate satisfies:

$$i_d = \max\{0, i_d^*\} \leq i_d^{PC}. \tag{52}$$

Equation (52) reflects the structure of deposit demand in (49). If competition is weak or the elasticity of deposit demand is low, banks would optimally choose $i_d^* < 0$. In that case however, buyers strictly prefer to hold reserves instead of deposits, since reserves for households have a return of zero. Consequently, banks set the deposit rate to the lowest feasible value that keeps buyers indifferent between deposits and reserves, namely, zero. Proposition 2 summarizes the welfare implications of the two banking regimes under imperfect competition:

Proposition 2. *Suppose holding money is costly, there is Cournot competition with N banks in the deposit market, and the banking system operates either under FB with $\alpha \in (0,1)$, or under NB with $\alpha = 1$. Then there exists a stationary equilibrium in which:*

a) $W_{FB} > W_{NB}$ and $i_{d,FB} = i_{d,FB}^* > i_{d,NB}$ if banking is sufficiently competitive, i.e., if $N > \bar{N}_{FB}$ where

$$\bar{N}_{FB} = \frac{\phi D}{(\partial \phi D / \partial i_d)} \frac{1}{i_d^{PC}} \tag{53}$$

is the level of competition that solves (51) at $i_d^* = 0$ for $\alpha \in (0,1)$.

b) $W_{FB} = W_{NB}$ and $i_{d,FB} = i_{d,NB} = 0$ if $N \leq \bar{N}_{FB}$.

Proposition 2 refines Proposition 1 by highlighting that, under imperfect competition, FB does not always yield higher welfare than NB. Although banks generate more income on the asset side under FB, because deposits finance higher-return loans rather than low-return reserves, they only pass this on to deposit holders if competition is sufficiently strong. In this case, the equilibrium deposit rate under FB, $i_{d,FB}^*$, is strictly higher than the deposit rate under NB, which is either the unconstrained imperfect competition rate $i_{d,NB}^*$, or zero. If competition is weak, i.e. $N \leq \bar{N}_{FB}$, banks would optimally choose $i_d^* \leq 0$ under FB. However, buyers then prefer to hold unremunerated reserves, which constrains banks to sed $i_d = 0$ in both regimes. As a result, FB and NB yield the same deposit and welfare in this case.

4 Quantitative Implications

This section calibrates the imperfect competition model to U.S. data in order to quantify the welfare costs of 100% reserve NB and to assess the implications of an interest-bearing CBDC.

4.1 Calibration

Following the literature, I parametrize the model with $u(q) = q^{1-\sigma}/(1-\sigma)$ and $U(x) = A\log(x)$. This implies the stationary allocation is given by:

$$x^* = A, \tag{54}$$

$$q_b = \left(\frac{1+i_d}{1+i}\right)^{1/\sigma}, \tag{55}$$

$$\phi D = (1-s)\frac{(1+i_d)^{\frac{1-\sigma}{\sigma}}}{(1+i)^{\frac{1}{\sigma}}}, \tag{56}$$

$$i_d = \max\left\{0, \frac{1+i_d^{PC}}{1+\frac{\sigma}{1-\sigma}\frac{1}{N}} - 1\right\}, \tag{57}$$

$$z = \frac{D}{D+x^*/\phi} = \frac{1}{1+\frac{B}{\phi D}}, \tag{58}$$

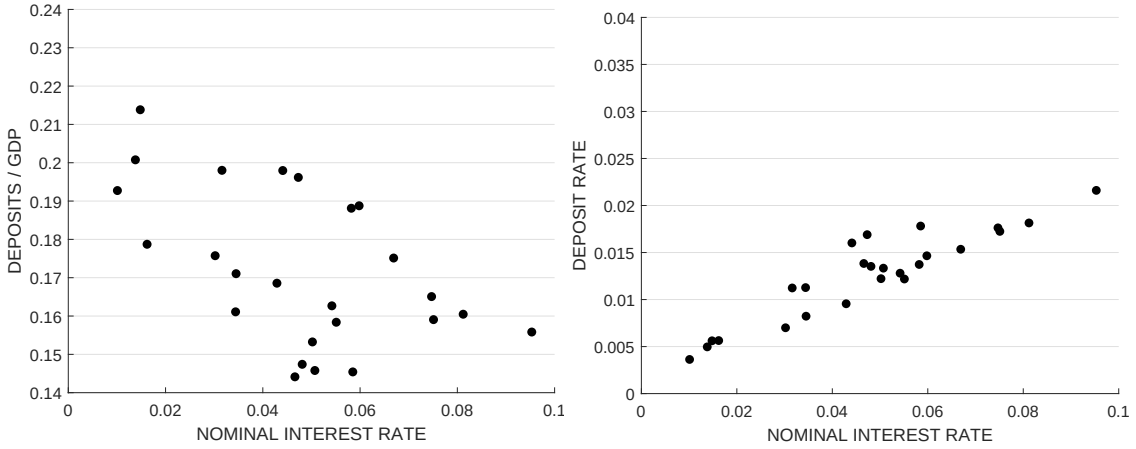
where z denotes the ratio of deposits to nominal GDP, i.e., aggregate real money demand or inverse velocity.[22]

Figure 3 plots U.S. money demand and the deposit rate from 1984 to 2008 against the 3-month T-bill rate as a measure of the nominal interest rate i. In what follows, I interpret each observed combination of real money demand and i, and the deposit rate and i, as a temporary steady state of the model, induced by exogenous changes in i implemented by the central bank.[23] Accordingly, equations (58) and (57) represent the model's theoretical counterparts to the data shown in Figure 3. As illustrated in the left panel of Figure 3, money demand decreases in i, consistent with the model's prediction that $\partial\phi D/\partial i < 0$. The right panel shows that the deposit rate rises with i, but significantly less than one-to-one due to the imperfect pass-through of policy rates to deposit rates. Capturing this wedge between the nominal interest rate and the deposit rate requires the imperfect competition framework introduced in Section 3.5.

[22] Total money balances held are given by total deposits D, which also equal nominal output in the FM. Nominal output in the SM is given by x^*/ϕ, so nominal GDP is $D + x^*/\phi$.

[23] For similar approaches in the New Monetarist literature see in particular Lagos and Wright (2005), Craig and Rocheteau (2008), and Berentsen et al. (2015). Lucas (2000), Lucas (2013), and Lucas and Nicolini (2015) make analogous arguments in a cash-in-advance setting.

Figure 3: U.S. Money Demand and Deposit Rates (1984–2008)



The calibration uses yearly U.S. data from 1984 to 2008.[24] Before the 1980s, the Glass-Steagall Act prohibited banks from paying interest on deposits. Only in 1982 were they permitted to issue interest-bearing money market deposit accounts (MMDA). Following Lucas and Nicolini (2015), I assume it took two years for banks to fully adopt this new account type, so the model mechanism becomes operative starting in 1984. The sample ends in 2008, as nominal interest rates approached zero thereafter.[25]

Table 1: Measurement of Model Variables

Model variable	**Data Measurement**
Deposits D	M1 excluding currency, plus MMDA
Reserves M	M0 excluding currency
Nominal interest rate i	3-month T-bill rate
Bank reserve rate i_r	Interest on bank reserves (0 in sample period)
Deposit rate i_d	Weighted average of M1 and MMDA deposit rates

Data sources: Federal Reserve Economic Data (FRED), Lucas and Nicolini (2015).

Table 1 summarizes how the model variables are measured. Following Chiu et al. (2023) and Altermatt and Wang (2024), deposits include MMDA, since MMDA perform a similar function to demand deposits. Moreover, money demand is not stable without this adjustment, as emphasized by Lucas and Nicolini (2015). Reserves are measured as M0 minus currency. Given that banks essentially held only required reserves αD during the period, the reserve requirement is calculated as $\alpha = M/D$. The nominal interest rate is measured by the 3-month T-bill rate. Since the Federal Reserve did not pay interest on reserves until October 2008, I set $i_r = 0$ throughout the sample. The deposit rate is constructed using data from Lucas and Nicolini (2015) on the average deposit rates for M1 deposits and MMDA, weighted by their respective shares in the aggregate.[26]

[24] Using HP-filtered data with $\lambda = 100$ yields practically identical parameter estimates.

[25] An interest rate of $i = 0$ violates the key assumption (5) that holding central bank money is costly, an assumption which underlies the binding reserve requirement and several other core mechanisms in the model.

[26] For the few years in which interest rate data for one of the two series is missing, I use a linear extrapolation.

With i, i_r and α taken directly from the data, the remaining parameters to calibrate are s, σ, A and N. These parameters are chosen to minimize the sum of squared residuals between the model-implied money demand and deposit rate, given by equations (54) to (58), and their empirical counterparts, z_t and i_{d_t}:[27]

$$\min_{s,\sigma,A,N} \quad SSR = \sum_{1984}^{2008} (z - z_t)^2 + \sum_{1984}^{2008} (i_d - i_{d_t})^2 . \tag{59}$$

To allow for changes in the degree of competition over the 25-year sample period, I also estimate a specification with two subperiods with flexible length, assigning a separate competition parameter N_1 for the first subperiod and N_2 to the second.

Table 2 reports the results of the calibration, and Figure 4 shows the corresponding model fit. Splitting the data between 1984–1990 and 1991–2008 yields the best fit. Notably, the estimation procedure in (59) only identifies the constant $k = A/(1-s)$, which enters the denominator of z, but not specific values for s and A.[28]

Table 2: Calibrated Parameters

$A/(1-s)$	σ	N_1	N_2	SSR
		1984–1990	1991–2008	
3.92	0.19	5	6	0.0085

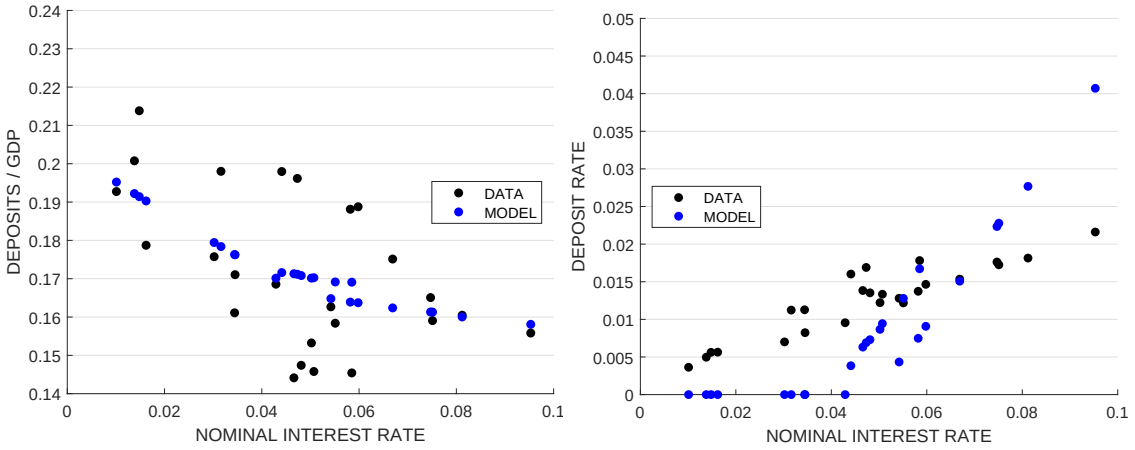
The parameters in Table 2 are broadly in line with those found in similar calibrations.[29] The result $N_2 > N_1$ suggests increasing competition over the sample period, which seems plausible given the financial deregulation that occurred during this time. The parameters from Table 2 with $N_2 = 6$ and $\alpha = 0.04$ (the average reserve requirement from 1984–2008), serve as the baseline calibration.

[27] Calibrating the model to match data moments or minimizing the sum of squared relative residuals yields very similar results.

[28] As explained below in footnote 30, the constant k is sufficient to compute welfare effects, so explicit values for s and A are not required.

[29] Lagos and Wright (2005) report $\{\sigma, k\} = \{0.16, 3.94\}$; Craig and Rocheteau (2008) find $\{\sigma, k\} = \{0.14, 3.64\}$; and Berentsen et al. (2015) estimate $\{\sigma, k\} = \{0.31, 4.48\}$ for similar model specifications. Chiu et al. (2023) find $\{\sigma, k, N\} = \{1.31, 3.73, 26\}$.

Figure 4: Model Fit



4.2 Welfare Costs of Narrow Banking

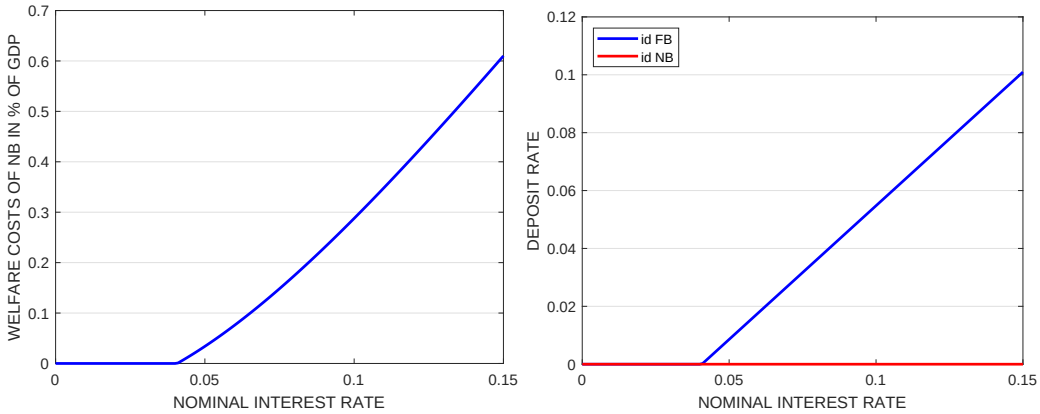
I first assess the welfare costs of a 100% reserve NB system. These are measured by calculating the fraction Δ_{FB} of expected steady-state consumption or GDP that households would be willing to give up under FB to attain the welfare level of a NB economy. In this benchmark, FM consumption $q_{b_{NB}}$ is given by (55) and (57) with $\alpha = 1$. The welfare cost of NB is then defined as $1 - \Delta_{FB}$.[30]

$$\begin{aligned} W_{FB}(\Delta_{FB}) &= (1-s)\frac{(q_b \cdot \Delta_{FB})^{1-\sigma}}{1-\sigma} - sq_s + A(\ln(A \cdot \Delta_{FB}) - 1) \\ &= (1-s)\frac{(q_{b_{NB}})^{1-\sigma}}{1-\sigma} - sq_{s_{NB}} + A(\ln(A) - 1) = W_{NB} \end{aligned} \tag{60}$$

We first consider the case in which the central bank does not pay interest on reserves, i.e., $i_r = 0$. This corresponds to a monetary system like that of the United States before 2008, when the Federal Reserve was not allowed to remunerate bank reserves. Figure 5 displays the welfare costs and deposit rates under FB and NB, as functions of the nominal interest rate i.

[30] Note that the individual values of A and s do not affect this welfare calculation; only the constant $k = A/(1-s)$ matters. This is due to the logarithmic specification of SM utility, which implies $\ln(A \cdot \Delta_{FB}) - \ln(A) = \ln(\Delta_{FB})$.

Figure 5: Wefare Costs of Narrow Banking with Unremunerated Reserves



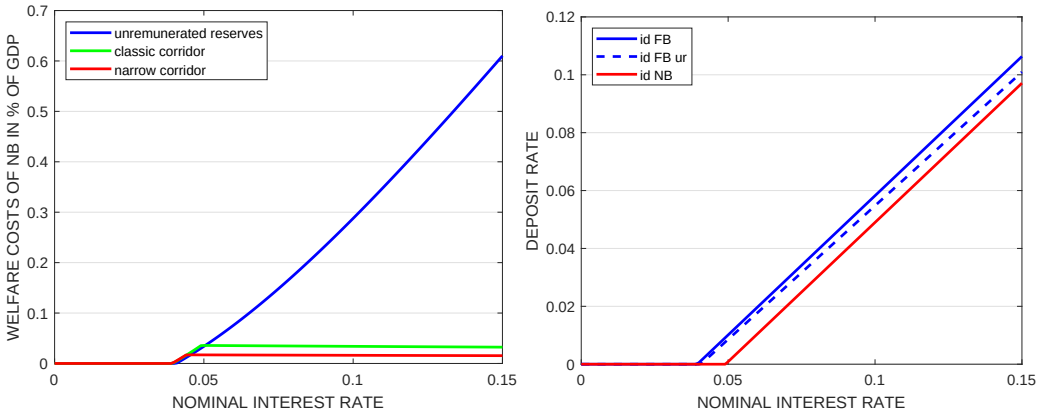
Notes: Baseline calibration with $i_r = 0$.

The left panel of Figure 5 shows that the welfare costs of NB increase with the nominal interest rate but remain modest except under high inflation. At a nominal interest rate of 4.8%, the U.S. average from 1984 to 2008, the welfare cost is approximately 0.03% of GDP. However, it rises to 0.29% and 0.61% of GDP at $i = 10\%$ and $i = 15\%$. As discussed above, these welfare costs arise from the differing abilities of the NB and FB regimes to generate interest on deposits, illustrated in the right panel of Figure 5. Under FB, the deposit rate i_d is zero at low nominal interest rates but rises linearly with i once i exceeds approximately 4%. This non-monotonicity, due to imperfect competition, was analyzed in Section 3.5 and captured in equation (57). When $i < 4\%$, banks would optimally set an unconstrained deposit rate $i_d^* < 0$. However, buyers would then switch to holding reserves and deposit demand would collapse. To avoid this, banks are forced to set $i_d = 0$. When $i > 4\%$, banks set a strictly positive deposit rate $i_d = i_d^* > 0$, which increases in the nominal interest rate. In this range, i_d^* under FB is always higher than the deposit rate under NB. In the NB regime, banks are always constrained by buyers' outside option: they cannot offer interest on deposits when reserves are unremunerated.

We now turn to the case in which the central bank remunerates bank reserves. Suppose the central bank operates a corridor system and remunerates reserves at a non-negative rate equal to a fixed spread μ below the nominal interest rate, i.e., $i_r = \max\{i - \mu, 0\}$. We consider two versions: a *classic corridor*, where $\mu = 100$ basis points, and a *narrow corridor*, where $\mu = 50$ basis points.[31]

[31] The classic (narrow) corridor corresponds to the average corridor width in the euro area before (after) the 2008 financial crisis.

Figure 6: Wefare Costs of Narrow Banking with Remunerated Reserves



Notes: Baseline calibration with $i_r = i - \mu$, where $\mu = 50(100)$ basis points in the narrow (classic) corridor. Deposit rates are shown for the classic corridor.

The left panel of Figure 6 shows that remunerating bank reserves substantially reduces the welfare costs of a 100% reserve NB system. For example, at $i = 15\%$, welfare costs fall from 0.6% of GDP with unremunerated reserves to 0.02% (0.03%) of GDP with reserve remuneration in the narrow (classic) corridor. The mechanism is illustrated in the right panel of 6, which shows that remunerating reserves significantly increases deposit rates under NB, shown here for the classic corridor, with an even stronger effect in the narrow corridor. Deposit rates under NB now exhibit behavior similar to those under FB: at low nominal interest rates, banks are constrained by the outside option of buyers, but above approximately $i = 5\%$, deposit rates track the bank reserve rate adjusted by the imperfect competition markup. As the corridor moves with the nominal rate, the deposit rate under NB rises proportionally. Deposit rates under FB are also higher with $i_r > 0$, but the effect is more muted. In FB systems, where deposits primarily finance loans, the reserve rate plays only a minor role in determining deposit rates. In contrast, under NB, where deposits exclusively finance reserves, the reserve rate is the sole source of income that banks can pass on to depositors. These results support the long-standing intuition of Friedman (1960), who advocated for a 100% reserve system with reserve remuneration to minimize welfare losses.

4.3 Effects of an Interest-Bearing CBDC

Finally, I briefly sketch the quantitative implications when households have access to an interest-bearing CBDC. Up to this point, I have assumed that reserves are remunerated for banks at rate $i_r \geq 0$, and for households at rate $i_z = 0$. This reflects the current monetary system in which only commercial banks have access to interest-bearing central bank money, while non-banks may only hold unremunerated currency. Now, I consider the case where reserves held by households

can also be remunerated at a rate $i_z \in [0, i_d^{PC}]$.[32] This can be interpreted as an interest-bearing retail CBDC. I will refer to i_z as the CBDC rate in what follows.

An interest-bearing CBDC changes the outside option for buyers relative to holding deposits. This affects their demand for deposits (49) as follows:

$$\phi D = \begin{cases} 0 & \text{if} \quad i_d < i_z, \\ (1-s)\frac{q_b u'(q_b)}{1+i_l} & \text{if} \quad i_d \geq i_z. \end{cases} \tag{61}$$

Buyers are willing to hold deposits only if the deposit rate is at least as high as the CBDC rate. Otherwise, they switch from holding deposits to holding CBDC, and the demand for deposits drops to zero. This puts pressure on banks' deposit rates, given by (52), which now take the form

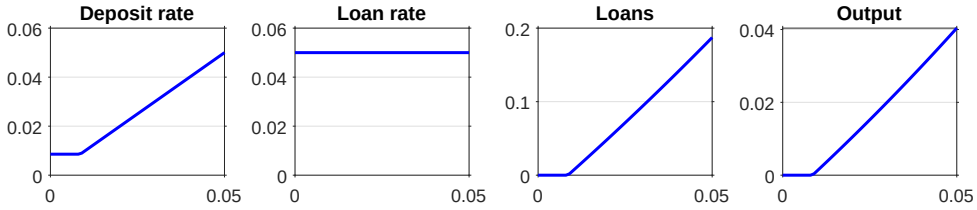
$$i_d = \max\{i_z, i_d^*\}. \tag{62}$$

When $i_z < i_d^*$, banks are unconstrained and set $i_d = i_d^*$, the profit-maximizing deposit rate under imperfect competition. However, if the CBDC rate rises above i_d^*, banks are forced to increase i_d one-to-one with i_z. This reflects the disciplining role of an interest-bearing CBDC, which offsets bank market power in the deposit market, as also emphasized by Chiu et al. (2023) and Andolfatto (2021).

Figure 7 illustrates the effects of the CBDC rate i_z on the deposit rate, the loan rate, real lending and output, assuming a nominal interest rate of 5% and unremunerated bank reserves. The first panel, which shows the deposit rate, illustrates equation (62). The flat segment corresponds to the unconstrained deposit rate under imperfect competition, i_d^*, which is approximately 0.9% in this calibration. For CBDC rates below 0.9%, the CBDC has no effect on the economy. However, once the CBDC rate exceeds this threshold, the deposit rate raises one-to-one with the CBDC rate. As shown in the second panel of Figure 7, this increase in bank funding costs does not raise the loan rate. In fact, it even increases or *crowds-in* lending and output, as seen in the last two panels. Higher deposit rates make deposits more attractive to buyers, which raises both deposit and loan demand, resulting in higher output and welfare.

[32] For the case where $i_z \in (i_d^{PC}, i)$, see appendix A.1. This scenario is less relevant because it requires unlikely parameter values, specifically i_r close to zero and i_z close to i. For example, in the calibration below with $i_r = 0\%$ and $i = 5\%$, we would need $i_z > 4.8\%$. When $i_z > i_d^{PC}$, the CBDC rate exceeds the maximum rate banks can offer on deposits. As a result, banks cannot compete with the CBDC, and buyers switch from deposits to CBDC in the FM. In this case, banks only intermediate CBDC from sellers to buyers and provide liquidity insurance through illiquid bonds. This corresponds to the BCW economy with remunerated central bank money, as outlined in Appendix A.1.

Figure 7: Effects of an Interest-Bearing CBDC



Notes: Baseline calibration with $i = 5\%$ and $i_r = 0$. Loans are given by (45) and output is defined as the sum of FM and SM output, $(1-s)q_b + A$. Both variables are expressed as changes relative to the equilibrium without a CBDC. Results with remunerated bank reserves are very similar.

These results are complementary to those in Chiu et al. (2023) and Andolfatto (2021). In Chiu et al. (2023), if the CBDC rate is high enough to constrain banks but below the deposit rate that banks would pay under perfect competition, then higher CBDC rates increase lending and reduce the loan rate. However, if the CBDC rate exceeds this threshold, it begins to *lower* lending and *raise* the loan rate. Intuitively, in the first regime, higher CBDC rates compel banks to increase deposit rates and thus counteract bank market power, as discussed above. This increases the supply of deposits from households, banks lend more and loan rates decrease, as in Andolfatto (2021), Section 4.1, where banks are subject to a reserve requirement. In the second regime, which is not analyzed in Andolfatto (2021), banks must raise loan rates as CBDC rates increase in order to break even, which reduces lending.

As highlighted in the introduction, a key difference from Chiu et al. (2023) and Andolfatto (2021) is that, in this paper, banks have economically viable funding alternatives to deposits. Since the models also differ along other dimensions,[33] it is difficult to determine to what extent the differing results are driven by this mechanism. Nevertheless, one robust distinction is that, in this model, a higher CBDC rate leaves the loan rate unchanged, whereas in the other papers, it affects the loan rate. This finding is consistent with the argument from Section 3.4 that sellers' demand for illiquid bonds insulates the loan rate from changes in the deposit rate.

5 Conclusion

This paper examines the costs of moving from the current FB system, where banks finance investment and lending using monetary liabilities such as deposits, to a 100% reserve NB system or a CBDC regime, both of which constrain banks' ability to create money. The analysis shows that, contrary to widespread concerns, if banks have economically viable alternatives to deposits for funding, constraining FB does not necessarily raise loan rates and thus reduce bank lending. Instead, the main welfare costs arise from lower interest paid on deposits, particularly in a competitive banking sector and under high inflation. In such environments, FB enables banks to better

[33] In particular, as discussed in the environment section, loan demand in Chiu et al. (2023) and Andolfatto (2021) originates from entrepreneurs, whereas in this paper it comes from households and is therefore closely linked to deposit demand.

compensate deposit holders for the inflation tax. However, the calibration suggests that these benefits are significant only at high inflation rates and decline sharply if the central bank remunerates reserves. To the extent that NB regulations reduce the financial instability associated with FB, enhancing financial stability through a NB system may thus not be so costly.

Declaration of generative AI and AI-assisted technologies in the writing process
During the preparation of this work the author used ChatGPT in order to improve language and readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

Lukas Altermatt and Zijian Wang. Oligopoly banking, risky investment, and monetary policy. *European Economic Review*, 164, 2024.

David Andolfatto. Assessing the impact of central bank digital currency on private banks. *The Economic Journal*, 131(634):525–540, 2021.

Raphael Auer, Jon Frost, Leonardo Gambacorta, Cyril Monnet, Tara Rice, and Hyun Song Shin. Central Bank Digital Currencies: Motives, Economic Implications, and the Research Frontier. *Annual Review of Economics*, 14(1):697–721, August 2022.

Philippe Bacchetta. The sovereign money initiative in Switzerland: an economic assessment. *Swiss Journal of Economics and Statistics*, 154(1):1–16, December 2018.

Aleksander Berentsen, Gabriele Camera, and Christopher Waller. Money, credit and banking. *Journal of Economic Theory*, 135(1):171–195, 2007.

Aleksander Berentsen, Samuel Huber, and Alessandro Marchesiani. Financial Innovations, Money Demand, and the Welfare Cost of Inflation. *Journal of Money, Credit and Banking*, 47(S2): 223–261, 2015.

Markus K. Brunnermeier and Dirk Niepelt. On the equivalence of private and public money. *Journal of Monetary Economics*, 106:27–41, 2019.

Stephen Ceccetti and Kermit Schoenholtz. Narrow banks wont stop bank runs. `https://www.moneyandbanking.com/commentary/2014/4/28/narrow-banks-wont-stop-bank-runs`, 04 2014.

Jonathan Chiu, Seyed Mohammadreza Davoodalhosseini, Janet Jiang, and Yu Zhu. Bank market power and central bank digital currency: Theory and quantitative assessment. *Journal of Political Economy*, 131(5):1213–1248, 2023.

Ben Craig and Guillaume Rocheteau. Inflation and Welfare: A Search Approach. *Journal of Money, Credit and Banking*, 40(1):89–119, 2008.

Douglas W. Diamond and Philip H. Dybvig. Bank Runs, Deposit Insurance, and Liquidity. *Journal of Political Economy*, 91(3):401–419, 1983.

Xavier Freixas and Jean-Charles Rochet. *Microeconomics of Banking, 2nd Edition*, volume 1. The MIT Press, 2008.

Milton Friedman. *A Program for Monetary Stability*. Fordham University Press, 1960.

Sebastian Infante, Kyungmin Kim, Anna Orlik, André F. Silva, and Robert Tetlow. Retail cbdc: Implications for banking and financial stability. *Annual Review of Financial Economics*, 16: 207–232, 2024.

Alberto Di Iorio, Anneke Kosse, and Ilaria Mattei. *Embracing diversity, advancing together - results of the 2023 BIS survey on central bank digital currencies and crypto*. Bank for International Settlements, 2024.

Timothy Jackson and George Pennacchi. How should governments create liquidity? *Journal of Monetary Economics*, 118(C):281–295, 2021.

Thomas Jordan. Why sovereign money would hurt switzerland, May 2018. URL `https://www.snb.ch/en/publications/communication/speeches/2018/ref_20180503_tjn`. Speech at the Swiss Institute of Banking and Finance at the University of St. Gallen.

Todd Keister and Daniel Sanches. Should Central Banks Issue Digital Currency? *The Review of Economic Studies*, 90(1):404–431, 03 2022.

Ricardo Lagos and Randall Wright. A unified framework for monetary theory and policy analysis. *Journal of Political Economy*, 113(3):463–484, 2005.

Jr. Lucas, Robert E. Glass-steagall: A requiem. *American Economic Review*, 103(3):43–47, 2013.

Robert E. Lucas and Juan Pablo Nicolini. On the stability of money demand. *Journal of Monetary Economics*, 73(C):48–65, 2015.

Robert E. Lucas, Jr. Inflation and welfare. *Econometrica*, 68(2):247–274, 2000.

George Pennacchi. Narrow banking. *Annual Review of Financial Economics*, 4(1):141–159, 2012.

Toni M. Whited, Yufeng Wu, and Kairong Xiao. Will Central Bank Digital Currency Disintermediate Banks? Working paper, 2023.

Appendix A

A.1 Baseline Economy with Illiquid Bank Liabilities (BCW economy)

Suppose the environment is identical to that in Section 2, but as in BCW, banks issue only illiquid liabilities (bonds) and the sole liquid asset in the economy is fiat money issued by the central bank (reserves), which is remunerated at rate $i_r = i_z \geq 0$ for both households and banks.[34]
Since banks do not issue deposits, they do not hold reserves in excess of the reserve requirement. Instead, as in BCW, they lend out deposited reserves to buyers. The bank's problem in (6) therefore simplifies to:

$$\max_{\{l,b\}\geq 0} \pi = l(1+i_l) - b(1+i_b)$$
$$s.t. \quad l = b.$$

As in the main text (see equation (8)), perfect competition implies $i_l = i_b$. Furthermore, since sellers still acquire bonds and buyers borrow from banks, the optimal reserve holdings of households in the SM are still given by equation (36). Thus, as in equation (42), we have $i_b = i_l = i$ in equilibrium, that is, both the bond rate and the loan rate equal the nominal interest rate.
The key difference relative to the main text is that buyers now use reserves, remunerated at rate i_r, rather than deposits, remunerated at $i_d \geq i_r$, in the FM. Consequently, the buyer's optimal FM consumption condition (20) is modified by replacing i_d with i_r, yielding:

$$u'(q_b) = p\phi\frac{1+i_l}{1+i_r}.$$

In equilibrium, this transforms equation (44) into:

$$\frac{u'(q_b)}{c'(\frac{1-s}{s}q_b)} = \frac{1+i}{1+i_r}. \tag{63}$$

This expression corresponds to the equilibrium FM consumption condition in BCW, given by equation (22) in their paper, when central bank money is remunerated. If, as in the baseline BCW model, central bank money is unremunerated cash (i.e., $i_r = 0$), the expressions are identical. Moreover, note that (63) also corresponds to the allocation in the baseline model under NB ($\alpha = 1$) when $i_d = i_r$, (44). This implies that the NB allocation in the baseline model is equivalent to the BCW allocation. Additionally, total loans in this economy are given by equation (45) with $\alpha = 1$ and $i_d = i_r$, which yields $\phi L = s(1-s)\frac{q_b c'(q_s)}{1+i_r}$. It follows that $\partial\phi L/\partial i_r > 0$ for $\sigma < 1$; that is, a higher remuneration of central bank money increases loans and intermediation.

[34] To facilitate comparison with the BCW model, I abstract from potential differences in reserve remuneration between households and banks in this section.

A.2 Baseline Economy with Liquid Bank Liabilities

Suppose the environment is identical to that in Section 2, except that banks can only issue liquid deposits as liabilities. The main implication of this restriction is that sellers can no longer hold bonds now and instead must hold deposits. As a result, their marginal value of holding reserves, given by (35), becomes $V_s'(m) = \phi(1 + i_d)$. Accordingly, the optimality condition for households' reserve holdings, given by (36), is modified to:

$$\phi = \beta s \phi_{+1}(1 + i_{d+1}) + \beta(1 - s)\phi_{+1}(1 + i_{l+1}),$$

which, under stationarity, simplifies to:

$$i = s i_d + (1 - s) i_l. \tag{64}$$

The deposit-pricing condition for banks, $i_d = \alpha i_r + (1 - \alpha) i_l$, from equation (7), still applies in this setting. Substituting this into the reserve holding condition yields the following expressions for the equilibrium loan and deposit rates:

$$i_l = \frac{i - s\alpha i_r}{1 - s\alpha} \quad , \quad i_d = \frac{\alpha i_r (1 - s) + (1 - \alpha) i}{1 - s\alpha}. \tag{65}$$

Observe that the loan rate now increases with the reserve requirement α, while the deposit rate decreases. This generates an additional spread between the loan and deposit rates, which translates into lower consumption (per equation (44)) and reduced welfare compared to the baseline model. Moreover, given that $i_l > i_d$, equation (64) shows that the preference shock s now influences interest rates and, through the spread $\frac{1+i_l}{1+i_d}$, also consumption and welfare. Thus, insurance against liquidity shocks is imperfect in this setting. One can further show that the spread $\frac{1+i_l}{1+i_d}$ increases with s, and that α marginally amplifies this effect. Therefore, in addition to the welfare loss from a lower return on deposits, a higher reserve requirement α now also worsens liquidity insurance.

Appendix B

B.1 proof of proposition 1

Proof. *a*): $\partial q_b/\partial s = 0$ (perfect insurance against the preference shock independent of α) is evident from (44); $\partial i_l/\partial \alpha = 0$ (unaffected loan rate) is evident from (42). For the connection between bank bond issuance and liquidity insurance see the text in Section 3.4.

b): $\partial i_d/\partial \alpha < 0$ is evident from (43) and $i > i_r$.

For the effect on FM consumption q_b, differentiate both sides of (44) with respect to i_d to get

$$\frac{\partial q_b}{\partial i_d} = -\frac{1+i}{(1+i_d)^2}\frac{1}{u''(q_b)} > 0, \tag{66}$$

where $\frac{\partial q_b}{\partial i_d} > 0$ follows from the strict concavity $u''(q_b) < 0$. Thus $\frac{\partial q_b}{\partial \alpha} = \frac{\partial q_b}{\partial i_d}\frac{\partial i_d}{\partial \alpha} < 0$.

For the effect on deposits, rewrite ϕD in (45) using (44) and differentiate with respect to i_d:

$$\phi D = (1-s)\frac{q_b u'(q_b)}{1+i}, \tag{67}$$

$$\frac{\partial \phi D}{\partial i_d} = (1-s)\frac{(\partial q_b/\partial i_d)\left(u'(q_b) + q_b u''(q_b)\right)}{1+i} = (1-s)\frac{(\partial q_b/\partial i_d)u'(q_b)\left(1-\sigma\right)}{1+i} > 0, \tag{68}$$

where the positive derivative follows from $\frac{\partial q_b}{\partial i_d} > 0$ and the assumption that the coefficient of relative risk aversion of buyers is below one, i.e. $\sigma = -\frac{q_b u''(q_b)}{u'(q_b)} < 1$. Thus $\frac{\partial \phi D}{\partial \alpha} = \frac{\partial \phi D}{\partial i_d}\frac{\partial i_d}{\partial \alpha} < 0$.

For the effect on loans, differentiate ϕL in (45) with respect to α as follows:

$$\frac{\partial \phi L}{\partial \alpha} = -(1-s)\phi D + (1-\alpha+\alpha s)\frac{\partial \phi D}{\partial \alpha} < 0, \tag{69}$$

which follows from $\frac{\partial \phi D}{\partial \alpha} < 0$.

For welfare, simplify W in (1) using the feasibility constraints and differentiate with respect to q_b:

$$W = (1-s)\left(u(q_b) - q_b\right) + U(x^*) - x^*, \tag{70}$$

$$\frac{\partial W}{\partial q_b} = (1-s)\left(u'(q_b) - 1\right) = (1-s)\left(\frac{1+i}{1+i_d} - 1\right) > 0. \tag{71}$$

Thus $\frac{\partial W}{\partial \alpha} = \frac{\partial W}{\partial q_b}\frac{\partial q_b}{\partial \alpha} < 0$.

c): see appendix A.1.

d): as $\alpha \to 0$, $i_d \to i$ from (43). Thus q_b in (44) $\to q_b^*$ in (4).

■

B.2 proof of proposition 2

Proof. Apart from the determination of the deposit rate in (52), the equilibrium conditions under imperfect competition are the same as under perfect competition. In particular, $i_l = i_b = i$ still holds, and q_b solves (44). This implies total real deposits, ϕD are independent of N and $i_d^{PC} = \alpha i_r + (1-\alpha)i$, thus $i_{d,FB}^{PC} > i_{d,NB}^{PC} = i_r$.

To proof proposition 2, note from (51) that if N increases from $\bar{N}_{FB}$, the right-hand side of (51) decreases. Thus (51) can only hold if the left-hand side decreases too which implies i_d^* must increase or $\partial i_d^*/\partial N > 0$. Therefore $i_d^* > 0$ if $N > \bar{N}_{FB}$. Conversely for $N < \bar{N}_{FB}$, $i_d^* < 0$ and thus $i_d = 0$. Now apply this to b), when $N \leq \bar{N}_{FB}$. Note that $i_{d,FB}^{PC} > i_{d,NB}^{PC}$ implies $\bar{N}_{FB} < \bar{N}_{NB}$ since $\bar{N}$ decreases in i_d^{PC} from (53). Thus if $N \leq \bar{N}_{FB}$ $N < \bar{N}_{NB}$ also holds, which implies $i_{d,NB}^* < 0$ and $i_{d,FB}^* < 0$ and thus $i_{d,FB} = i_{d,NB} = 0$.
If $N > \bar{N}_{FB}$ we have two cases: Either $\bar{N}_{FB} < N < \bar{N}_{NB}$ or $N > \bar{N}_{NB}$. The first case implies $i_{d,NB}^* < 0$ and thus $i_{d,NB} = 0$ and $i_{d,FB} = i_{d,FB}^* > 0$. Thus $i_{d,FB} > i_{d,NB}$ holds.
In the second case both $i_{d,FB} = i_{d,FB}^* > 0$ and $i_{d,NB} = i_{d,NB}^* > 0$. To show $i_{d,FB}^* > i_{d,NB}^*$ reformulate (51) as

$$i_d^{PC} = i_d + \frac{i_d}{\varepsilon_D(i_d)}\frac{1}{N} = i_d + \frac{\phi D}{(\partial \phi D/\partial i_d)}\frac{1}{N} = i_d + m(i_d) = RHS(i_d), \tag{72}$$

where $m(i_d)$ denotes the markup between the perfect and the imperfect competition rate, and $RHS(i_d)$ is the right-hand side of (72). Now use (66) to (68) from the proof above, (44) and the definition of σ to rewrite $m(i_d)$ as:

$$m(i_d) = \frac{q_b}{1-\sigma}\frac{1}{(\partial \phi D/\partial i_d)}\frac{1}{N} = \frac{q_b}{1-\sigma}\frac{-u''(q_b)}{u'(q_b)}(1+i_d)\frac{1}{N} = \frac{\sigma}{(1-\sigma)}(1+i_d)\frac{1}{N}. \tag{73}$$

Thus $RHS(i_d)$ linearly increases in i_d. Ergo $RHS(i_{d,FB}^*) = i_{d,FB}^{PC} > RHS(i_{d,NB}^*) = i_{d,NB}^{PC}$ implies $i_{d,FB}^* > i_{d,NB}^*$.

■